A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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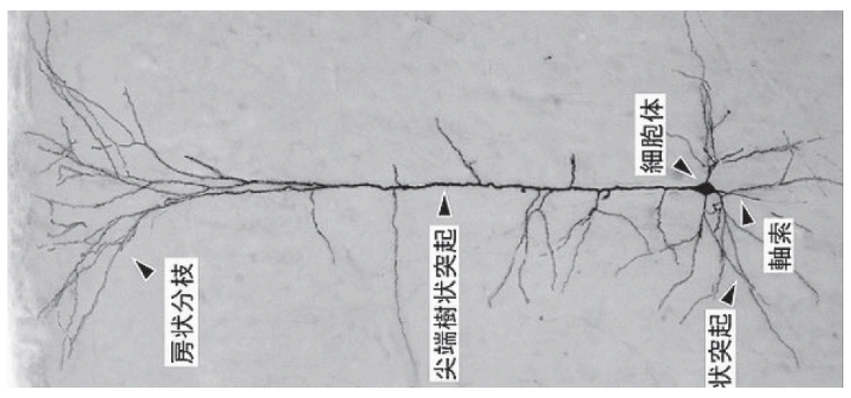
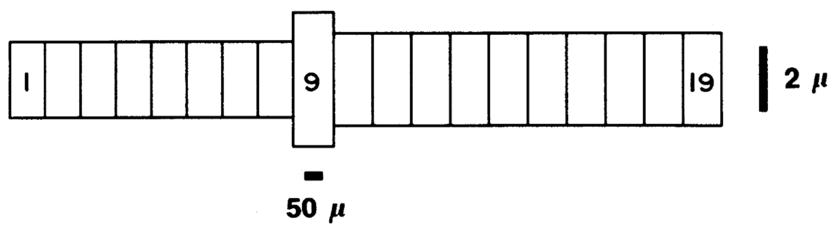
Introduction

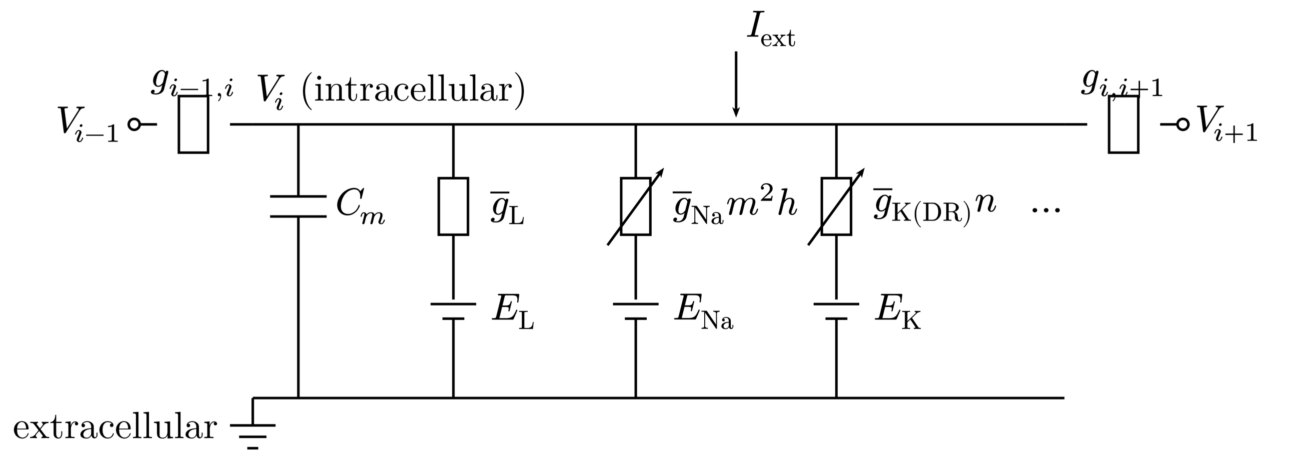


CA3 pyramidal neuron [1]



Modelled as **19 compartments** [2]

A multi-compartment model represents the **spatial morphology** as connected compartments.



Equivalent circuit of compartment i

Each compartment is an **RC circuit**: capacitance, **variable** ionic conductances ($\cdots = g_{Ca}, g_{K(A)}, g_{K(AHP)}, g_{K(C)}$), and axial coupling $g_{i,i\pm1}$.

Conductances depend on **gate variables**, which obey ODEs with **rate functions** α, β:

$$I_{Na} = \bar{g}_{Na}\, m^2 h\, (V - E_{Na})$$

$$\frac{dm}{dt} = \alpha_m(V)(1-m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1-u)/(\exp((13.1-u)/4)-1)$$

Na current, its gate m, and one rate function ($u = V - E_L$)

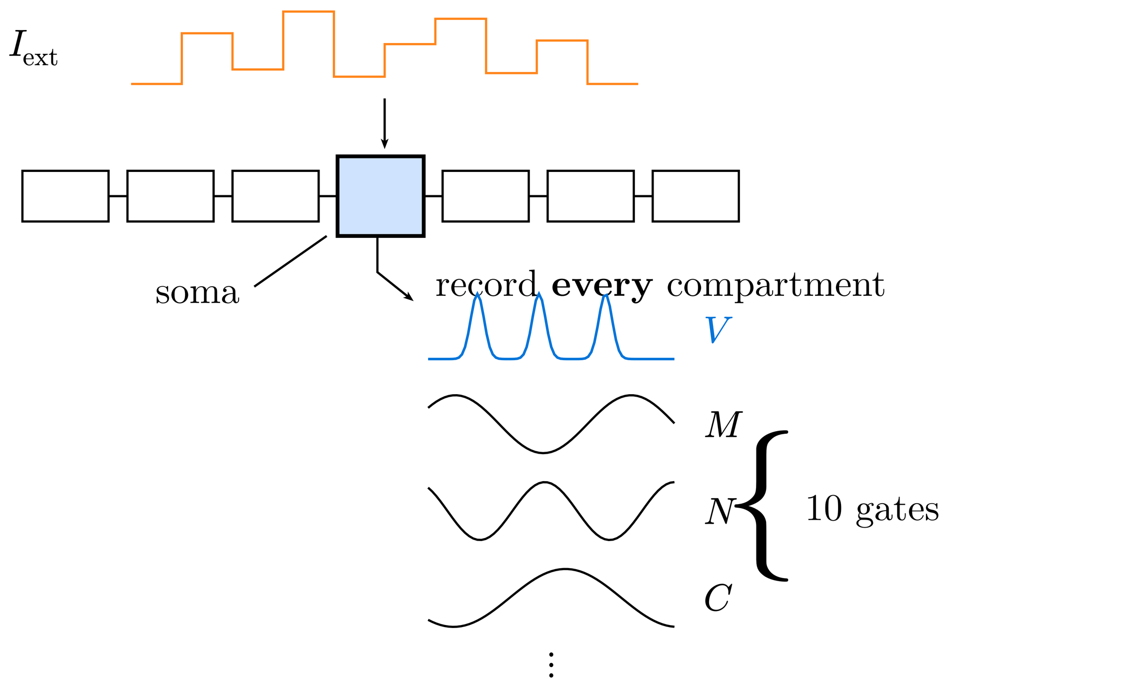
→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

Goal

Reproduce the membrane-potential response with **fewer state variables**.

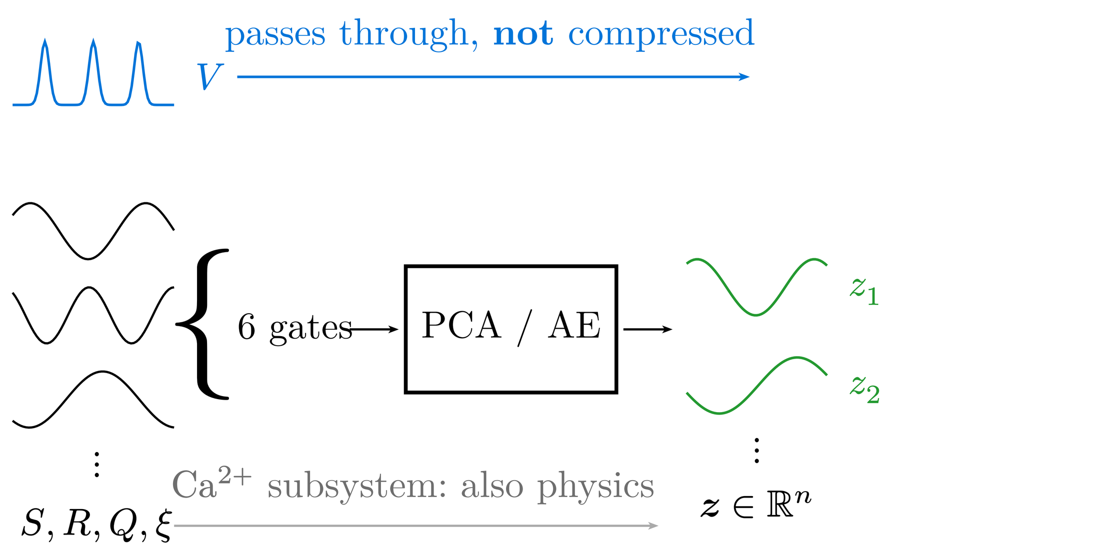
Methods

① Collect the training data



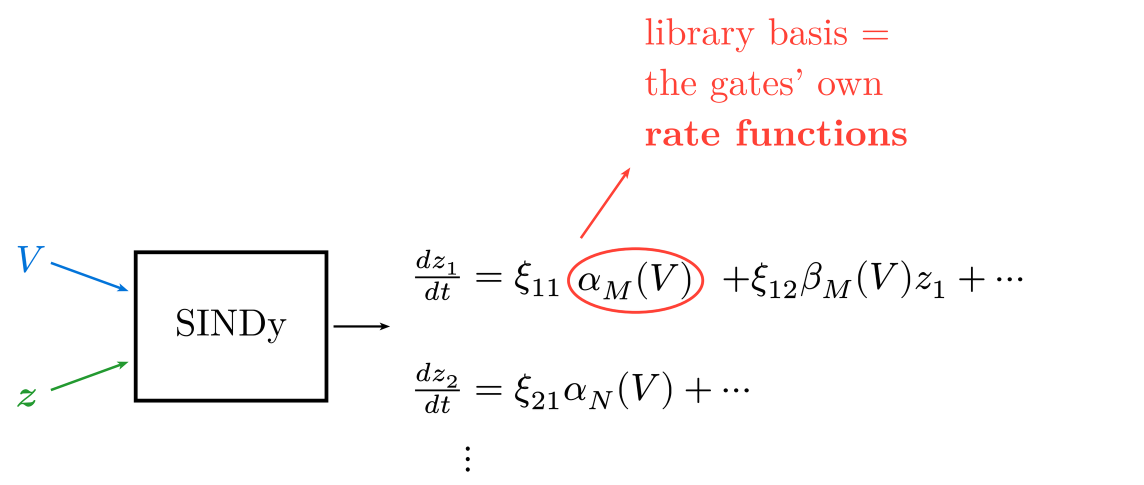
Inject a **random pulse train** at the **soma** of the Traub 19-compartment cell; record V and the **10 gates** of **all 19 compartments**.

② Compress *only* the gates



The **6 voltage-dependent gates** ride on a **low-dimensional manifold** → $z \in \mathbb{R}^n$, $n = 5$. V and the Ca^{2+} subsystem **stay uncompressed**.

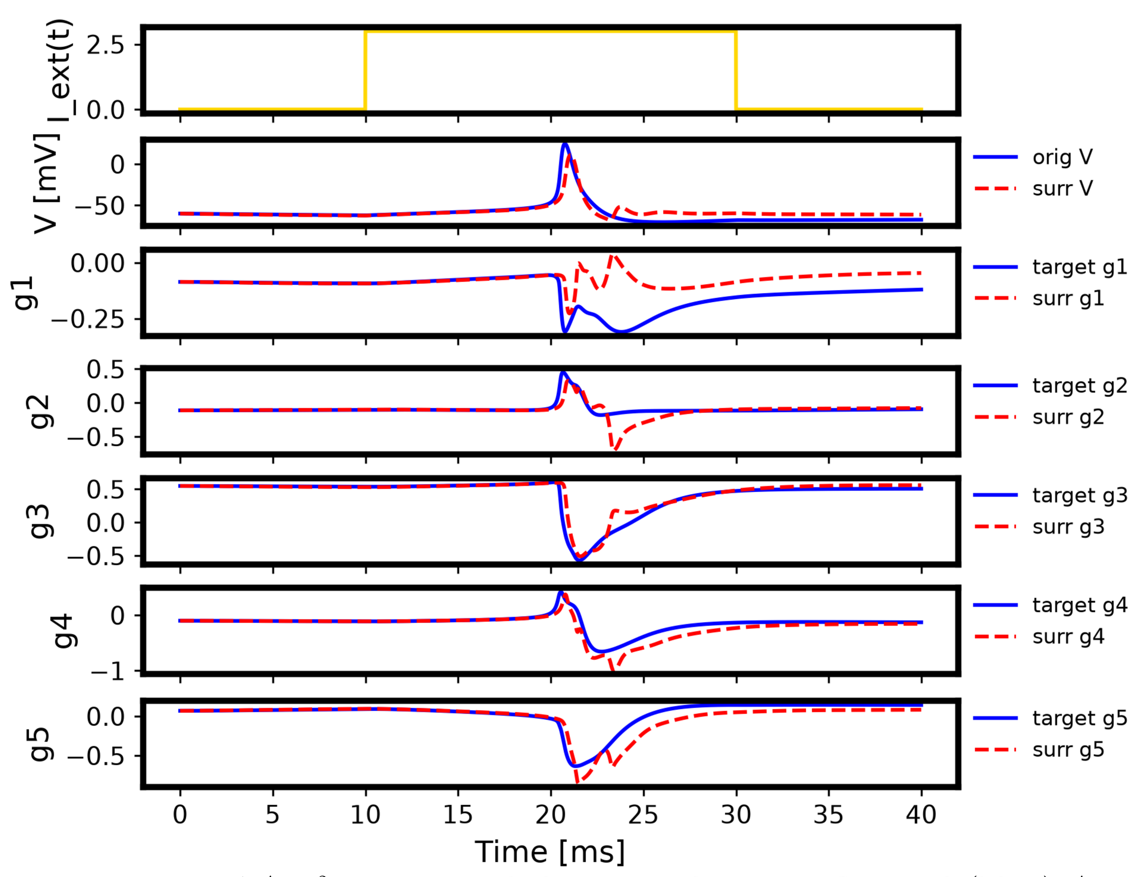
③ Identify the latent ODEs



SINDy [3] fits **only** $\dot{z}$; $\dot{V}$ and the Ca^{2+} subsystem keep the **original physics**. The library is **physics-informed** — the gates' own $\alpha(V), \beta(V)$, not last year's **41-term** generic one.

Results and Discussion

① Single action potential



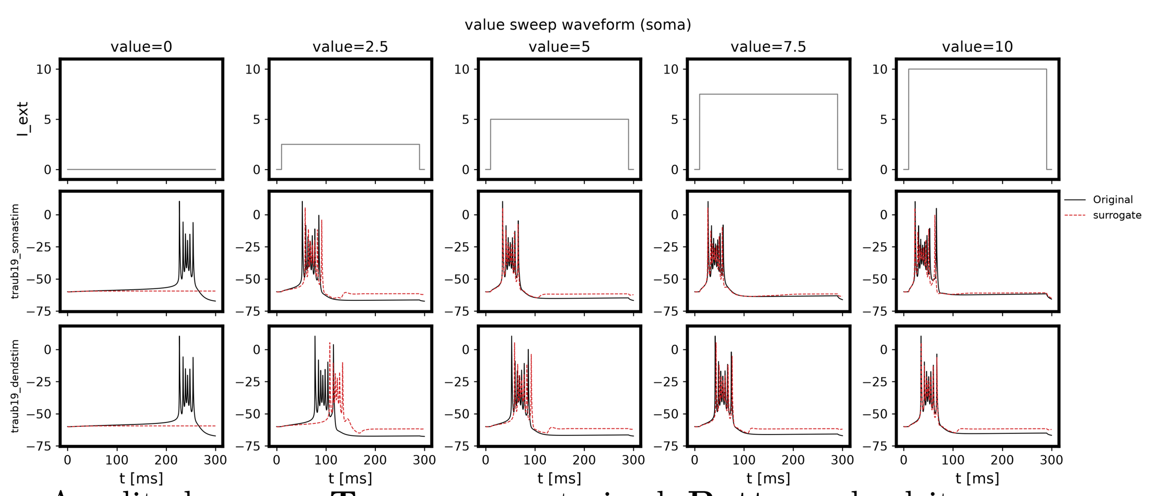
20 ms, 3 μ A/cm² step: V and the **5 AE latents**. Original (blue) / surrogate (red).

Waveform match

- RMSE **7.3 mV**, shape corr. **0.999**
- spike **1 → 1**, latency err **0.3 ms**

- **Gap**: peak **13 mV low**.

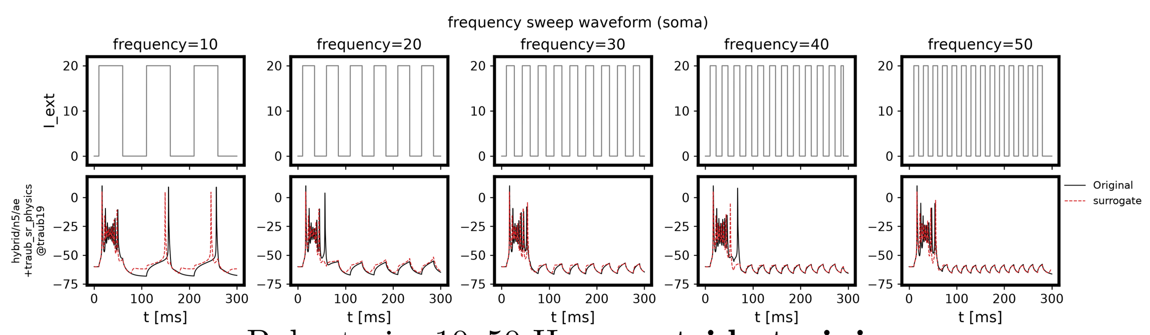
② New stimulus site



Amplitude sweep. **Top**: soma, as trained. **Bottom**: dendrite, **unseen**.

- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

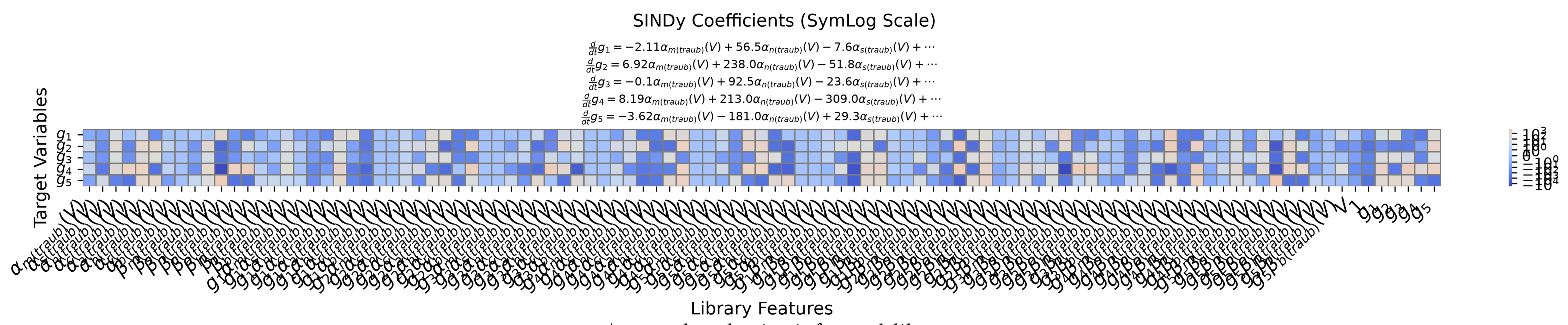
③ Unseen periodic drive



Pulse train, 10–50 Hz — **outside training**.

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

④ Identified latent equations



ξ over the physics-informed library.

- **79.6% non-zero** — accurate but **not sparse**: cost **rises** (exp 19 → 121).

Conclusion

- **All 19 compartments** replaced, no divergence; AP **shape corr. 0.999**.
- Transfers to an **unseen site and frequency**.
- Compression stops at **6 → 5** and ξ stays dense.

→ **Future**: sparsify ξ, push n down.

Code — https://github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] "錐体細胞." Accessed: Dec. 08, 2025. [Online]. Available: https://doi.org/10.14931/bsd.2091

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.